



SCALING FINNISH SAAS STARTUPS: AN EMPIRICAL STUDY

Lappeenranta–Lahti University of Technology LUT

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ABSTRACT

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Scaling is a critical but challenging phase for software startups, with most failing at this stage. While the topic has been studied more systematically, research focuses on large and capital-intensive ecosystems. At the same time, the Nordic region, including Finland, has become the fastest-growing software export hub in Europe. This qualitative study explores how Finnish SaaS startups scale their operations from a small but digitally mature market. The data consists of six in-depth interviews, and the analysis was conducted using the Gioia methodology. The results are organized into three core dimensions: business, organizational, and technological. Based on the findings, Finnish companies approach scaling by combining international orientation, lean data-driven decision-making and resource efficiency. Some proceed step by step using cash flow, while others rely on external funding. They leverage Inbound and PLG strategies for global reach, establish hubs to key markets and build trust based distributed engineer teams. Product development focuses on the most critical growth constraints and the most value-producing features, and the companies maintain one unified product without customer-specific modifications. Finland, if not skipped entirely, is used as a test market for globally scalable products. This study contributes to the understanding of SaaS scaling practices, specifically in smaller market contexts. Future research can (1) further deepen the understanding of Finnish SaaS scaling by studying failed SaaS startups, and (2) by exploring other emerging software export hubs to identify their own scaling paths.

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Skaalautuminen on ohjelmistoalan kasvuyrityksille kriittinen vaihe, mutta suurin osa epäonnistuu juuri tässä kohtaa. Kasvava nykytutkimus keskittyy suuriin ja pääomavaltaisiin ekosysteemeihin. Samaan aikaan Pohjoismaista, Suomi mukaan lukien, on tullut Euroopan nopeimmin kasvava ohjelmistoviennin keskus. Tämä laadullinen tutkimus tarkastelee, miten suomalaiset SaaS-kasvuyritykset skaalaavat toimintaansa pienestä mutta digitaalisesti kypsästä markkinasta käsin. Aineisto koostuu kuudesta syvähaastattelusta, ja analyysi on toteutettu Gioia-menetelmällä. Tulokset jäsennetään kolmen keskeisen ulottuvuuden mukaan: liiketoiminta, organisaatio ja teknologia. Tulosten perusteella suomalaiset SaaS-kasvuyritykset lähestyvät skaalausta yhdistämällä kansainvälisen suuntautumisen resurssitehokkaaseen ja datavetoiseen nopeaan päätöksentekoon. Osa etenee vaiheittain kassavirralla, osa ulkopuolisen rahoituksen avulla, yleensä inbound- ja PLG-strategioin, sekä perustamalla keskuksia aavainmarkkinoille. Samanaikaisesti he hyödyntävät globaalia ohjelmointiosaajien markkinaa ja rakentavat luottamukseen pohjautuvia hajautettuja tiimejä. Tuotekehityksessä keskitytään kriittisimpiin kasvun esteisiin ja arvoa tuottavimpiin ominaisuuksiin, ylläpitäen yhtä yhtenäistä tuotetta ilman asiakaskohtaisia muokkauksia. Suomea hyödynnetään testialustana globaalisti skaalautuville tuotteille, jos ei ohiteta kokonaan. Tutkimus täydentää ymmärrystä SaaS-yritysten skaalauskäytännöistä erityisesti pienempien markkinoiden kontekstissa. Jatkotutkimuksessa (1) tätä kontekstia voidaan edelleen täydentää tutkimalla epäonnistuneita SaaS-kasvuyrityksiä, sekä (2) tutkia muita nousevia ohjelmistovientikeskuksia niiden skaalautumispolkujen selvittämiseksi.

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SYMBOLS AND ABBREVIATIONS

Abbreviations

API	Application Programming Interface
ARR	Annual Recurring Revenue
CAC	Customer Acquisition Cost
CEO	Chief Executive Officer
CI/CD	Continuous Integration / Continuous Deployment
GTM	Go-To-Market
ICP	Ideal Customer Profile
IEEE	Institute of Electrical and Electronics Engineers
IRR	Internal Rate of Return
IPO	Initial Public Offering
KPIs	Key Performance Indicators
LTV	Customer Lifetime Value
LUT	Lappeenranta-Lahti University of Technology LUT
MRBS	Massive and Rapid Business Scaling
MRQ	Main Research Question
MRR	Monthly Recurring Revenue
MVP	Minimum Viable Product
NRR	Net Revenue Retention
PLG	Product-Led Growth
PLS	Product-Led Sales
PMF	Product-Market Fit
ROI	Return on Investment
SaaS	Software as a Service
SME	Small and Medium Enterprise(s)
TAM	Total Addressable Market
VC	Venture Capital

Table of contents

Abstract

(Acknowledgements)

(Symbols and abbreviations)

1	Introduction	11
2	Literature Review	13
2.1	SaaS Software Startups	13
2.1.1	Startup Definitions and Characteristics	13
2.1.2	SaaS Business Model.....	15
2.2	Concept and Importance of Scaling	16
2.2.1	Defining Scaling, Scale-Up & Scalability	17
2.2.2	Timing & Readiness	19
2.3	Business Aspects.....	20
2.3.1	Funding Strategies	20
2.3.2	Sales, Marketing & Expansion	22
2.3.3	Growth Road Mapping, Experimentation and Business Model Evolution	24
2.4	Organizational and Process Aspects of Scaling	25
2.4.1	Building Organizational Capacity, Agility & Talent.....	25
2.4.2	Operational Excellence & Process Innovation	26
2.5	Technology Aspects	27
2.5.1	Software Engineering	27
2.5.2	Technical Debt Management.....	28
2.5.3	Scalable Technologies & Infrastructure	29
2.6	Synthesis, Research Gaps & Research Questions.....	31
2.6.1	Synthesis	32
2.6.2	Summary.....	35
2.6.3	Research Gaps.....	36
3	Methodology	37
3.1	Research Objective.....	37

3.2	Research Design.....	38
3.2.1	Overall Approach.....	38
3.2.2	The Gioia Methodology Framework	39
3.3	Data Collection and Participants.....	39
3.4	Data Preparation and Data Analysis	40
3.5	Research Quality and Trustworthiness.....	42
3.6	Ethical considerations	42
3.7	Limitations	43
4	Findings	44
4.1	Scaling Prerequisites	44
4.1.1	Global Ambition	45
4.1.2	Product-Market Fit & Metrics	46
4.1.3	Funding	47
4.2	Business	48
4.2.1	Internationalization	48
4.2.2	Marketing & Sales Strategies	49
4.3	Organizational & Process.....	50
4.3.1	Lean Planning	50
4.3.2	Experimentation.....	51
4.3.3	People Practices	52
4.4	Technology.....	53
4.4.1	Single Product Focus	53
4.4.2	Technical Debt Management Strategies	54
5	Discussion	55
5.1	How do Finnish SaaS startups develop and apply internal strategies during scaling?	56
5.2	How do external conditions shape the scaling approaches of Finnish SaaS startups?	57
6	Conclusions	59
	References.....	61

Appendices

Appendix A. Gioia Data Structure

1 Introduction

Software startups deliver much of today's software innovation (Nguyen-Duc et al., 2021), often with the software-as-a-service (SaaS) model (Saltan and Smolander, 2021). These startups, aiming to disrupt the industry, navigate through chaos with ad-hoc practices, resource, skill, and time constraints, and generate technical debt while progressing between distinct life-cycle stages with evolving goals: 1st building the MVP in inception; 2nd ensuring product maintainability in stabilization; 3rd scaling to the desired market share and growth rate in growth; and 4th preserving market share and optimizing operations in maturity. (Klotins et al., 2019)

Of all the phases, 'Scaling,' the time-limited phase of extremely rapid growth (Bohan et al., 2024), where startups undergo an organizational transformation process to achieve returns to scale with digital technologies, emerges as the most challenging for these ventures (Coviello et al., 2024). This complex process which requires careful management and tight interplay between business model, technology architecture and organization design is in which during most of the startups fail, however success in scaling predicts long term viability, making it critical to success with. (Coviello et al., 2024; Lee and Kim, 2024; Sanasi et al., 2023). For software startups focused on building and marketing their software-intensive products, the scaling journey introduces unique challenges, the core challenge being the product as the user base and requirements grow rapidly and technical debt builds up (Klotins et al., 2018; Klotins et al., 2019; Wang, 2016).

Current research of software startups during scaling strategies and activities are limited, the current research with broader digital startup context has identified and aimed to model the timing, prerequisites and key strategies and activities such as business-model innovation, expansion, experimentation, culture and organization transformation and technologies to achieve operational excellence (Lange et al., 2023; Mula et al., 2024; Rayport et al., 2023; Sanasi et al., 2023). While some research has focused purely on technical aspects, e.g. scalable architecture (Ajiga et al., 2024).

'Scaling' in the business context has recently gained much deserved attention from practitioners, policy makers and academics (Bohan et al., 2024), literature of scale-ups still

limited Mula et al., 2024). E.g. Lange et al. 2023; Lee and Kim, 2024; Mula et al., 2024; focus on broader digital startup context from established markets. At the same time, according to McKinsey (Bjørndalen et al, 2024): the Nordic region, including Finland, has emerged as Europe's most export-intensive and fastest-growing software hub. Finnish SaaS startups operate in this unique environment, characterized by a small home market, high digital maturity and a strong base of engineering talent, thus calling for research that takes this unique context into account (Bjørndalen et al, 2024).

Therefore, the aim of this study is to build an empirical understanding of how Finnish SaaS startups approach scaling through the experiences of Finnish SaaS founders. To achieve this aim, the study is guided by the following main research question (MRQ): How do Finnish SaaS startups approach scaling?

This study used a qualitative, inductive methodology. Six founders and leaders of distinct active Finnish SaaS startups, from various employee ranges (Crunchdata, 2025), shared their experience of their company scaling via semi structured interviews. The interview data was analyzed using the Gioia method (Gioia et al., 2013) to identify emergent themes and understand scaling in this context.

The findings of this study present emerged strategies and practices from the Finnish SaaS scaling journey, that both align with existing empirical understanding, well as underscore the unique context specific characteristics.

Chapter 2 reviews the relevant literature on startup scaling, combining it with the literature on software startups (e.g. characteristics, challenges and life cycle) and the SaaS business model. Chapter 3 details the research methodology, outlining the research design, data collection and the framework used. Chapter 4 presents the findings from the interview data, structured by the emergent themes and dimensions. Chapter 5 discusses these findings. Chapter 6 concludes the study by summarizing the findings, discussing the implications for theory and practice, noting limitations and suggesting future research.

2 Literature Review

This literature review is structured around three core dimensions: business, technology, and organizational. It draws the classification from SaaS adoption research that highlights business and technology and engineering aspects (Saltan & Seffah, 2018) and adapts this perspective to the context of startup scaling described as organizational process involving the interplay of internal transformation, technical architecture, and evolving business models (Bohan et al., 2024; Coviello et al., 2024). Since there are few direct studies on how SaaS startups scale, this review uses related research to build a solid foundation.

Sources were gathered using Google Scholar, with access through LUT University's library tools, focusing on publications from 2010 to 2024. Keywords included 'software startup scaling', 'software startups', and 'SaaS scaling'. Alongside peer-reviewed studies, selected industry reports and practitioner sources were included.

To structure the review and support the development of the research questions, the chapter is organised into five main sections: Section 2.2 defines the concept of scaling and discusses its timing and readiness. Sections 2.3, 2.4, and 2.5 cover the business, organizational, and technical dimensions of scaling. Section 2.6 concludes with a synthesis of key insights and identifies research gaps that guide the development of this study's research questions.

2.1 SaaS Software Startups

Software startups' main focus is building and marketing software-intensive products, often using a SaaS model (Klotins, 2019; Saltan and Smolander, 2021). This section reviews the definitions, characteristics, and challenges of software startups and SaaS.

2.1.1 Startup Definitions and Characteristics

Startup, according to two widely recognized definitions from both academia and industry, by Steve Blank and Eric Ries:

- **Blank:** ‘A temporary organization that exists to find a scalable, repeatable, and profitable business model’ (Blank, 2010).
- **Ries:** ‘A human institution designed to create new products and/or services under conditions of extreme uncertainty’ (Ries, 2007).

These two definitions are widely accepted and are complementary. While Blank’s definition is more goal-oriented, Ries’ definition describes the operating environment.

Software startup, expanding to Ries (2007) and Blank (2010) is such a temporary venture searching for their scalable business models in the extreme uncertainty, focus marketing and building software intensive products (Klotins et al., 2021; Unterkalmsteiner et al., 2016). Software startups are characterized across the literature as newly established, small, and innovative, with high growth goals and high failure rates. They operate with scarce resources, such as people, skills, and funding. They often have no prior or only limited operating history, resulting in immature processes. They function in dynamic environments where speed is a priority (Boch et al., 2013; Unterkalmsteiner et al., 2016; Klotins et al., 2021; Wang et al., 2016). Due to limited operating history, limited funding, small teams, and lack of domain expertise, startups adopt fast, iterative development approaches as the startups must quickly test product ideas, gather feedback, and decide whether to pivot or continue (Boch et al., 2013; Wang et al., 2016; Klotins et al., 2019; Unterkalmsteiner et al., 2016). Klotins et al. (2019) describe the iterative release of a minimum viable product (MVP) to gather early customer feedback. Wang et al. (2016) show empirically that “building the product” consistently ranks as a top challenge for startups in multiple life-cycle stages. Unterkalmsteiner et al. (2016) note that short feedback loops are key to surviving volatile market shifts. Due to ad-hoc engineering practices and shortcuts in product development to reduce time-to-market, startups accumulate technical debt (Klotins et al., 2018). Balancing quality and speed, along with securing funding, are central challenges for startups (Unterkalmsteiner et al., 2016; Klotins et al., 2021; Wang et al., 2016).

2.1.2 SaaS Business Model

Software-as-a-Service (SaaS) is a delivery model in which software application access is provided to the end user over the internet, and the application is managed and configured on behalf of the user (Mell and Grance, 2011). SaaS represents the biggest segment of the public cloud computing market and has become the dominant model for software licensing and delivery worldwide (Saltan and Smolander, 2021). In the SaaS model, software is delivered via subscription and accessed remotely by users, eliminating the need for local installation on personal computers or servers (Saltan and Smolander, 2021).

SaaS applications are typically deployed in a multitenancy architecture leveraging on-demand cloud computing. In a multi-tenant architecture, a single application instance is deployed with a single database, which is ‘shared’ by multiple users (tenants), allowing them to configure the application as if they were in a dedicated environment. Multi-tenancy Maximizes hardware utilization and simplifies deployment, which leads to lower costs (Bezemer and Zaidman, 2010).

Luoma et al. (2012) offer a solid classification of SaaS business model variations (‘Pure-Play’, ‘Enterprise’, ‘Self-Service’) and their evolution over time, implying the progression from less profitable, high-automation models to more mature, high-touch enterprise models.

The SaaS business model centers around a simple, standardized offering with little or no services. It has an efficient sales model for targeting SME customers of all levels, with a focus on customer acquisition and retention, automating the processes, delivery, offering, and support, and relying on scalable infrastructure for returns to scale. The SaaS model includes usage-based pricing in which transactions are small. It involves high up-front investment and development costs, but low cost per customer. The goal is to grow a large user base to increase recurring revenue (Luoma et al., 2012). Highlighted by both Cespedes and van der Kooij (2023) and Floerecke (2018), a key factor not listed by Luoma et al. in the SaaS business is investing in a customer success function. This is due to the SaaS model’s reliance on slowly building recurring revenue. SaaS companies should be able to acquire new customers, retain them, and upsell (Skok, n.d.). A key to SaaS model sustainability lies in measuring the right metrics.

Luoma et al. (2012) outline three SaaS business model variations: ‘Enterprise SaaS’, ‘Pure-Play SaaS’, and ‘Self-Service SaaS’. The ‘Enterprise SaaS’ model leans on ‘high-touch’ and high-value deals with longer sales cycles, targeting top executives and IT managers of large enterprises, with a focus on personal relationships and trust-building. It can require manual onboarding, integration, and custom contracts. ‘Pure-Play SaaS’ refers to horizontal, standard web apps. This business model leverages online channels, marketing, and push-oriented inbound sales. It targets SME end users and middle management, relies on heavy automation, and aims for minimal costs. ‘Self-Service SaaS’ relies on a simple application, freemium models, ads, or small recurring fees. It is fully automated. A strong landing page is critical, and the marginal cost is close to zero.

As the SaaS business model relies on recurring revenue, as emphasized by Luoma et al. (2012), succeeding in SaaS requires companies not only to acquire customers but also to retain them for as long as possible to maximise customer lifetime value (LTV). Due to this, focus on retention and churn is important. David Skok discusses the importance of unit economics in measuring a startup’s viability (Skok, n.d.), which is also supported by Cespedes and van der Kooij (2023).

According to Skok, it is important to track Customer Lifetime Value (LTV) and Customer Acquisition Cost (CAC) ratio, which should ideally aim for a 3:1. Additionally, companies should focus on keeping churn low. This reflects whether the recurring revenue stream can sustainably cover acquisition costs. Another key metric for business health is Net Revenue Retention (NRR), which measures how much revenue is retained and how much it grows over time by adding expansion revenue and subtracting contraction revenue, divided by the starting revenue (Cespedes and van der Kooij, 2023).

2.2 Concept and Importance of Scaling

The lifecycle of a software startup is identified as having various distinct stages: inception, stabilization, growth, and maturity, as outlined by Klotins et al. (2019) based on Crowne (2002). Each of these stages has unique objectives, requires specific competencies, and introduces its own set of challenges (Klotins et al., 2019; Get al., 2019; Wang et al., 2016;

Nguyen-Duc et al., 2021; Melegati et al., 2024). The stage of ‘growth’, as described by Klotins et al. (2019), has the goal of scaling up the business to achieve the target growth rate and market share. This ‘scaling phase’ is the most critical stage, during which most startups fail (Sanasi et al., 2023; Lee and Kim, 2024).

2.2.1 Defining Scaling, Scale-Up & Scalability

The current startup scaling literature, in the context of digital and technology startups, discusses scaling as efficient, rapid growth of the customer base and/or returns to scale, a stage of high growth, often referred to as ‘special growth’ (Coviello et al., 2024; Mula et al., 2024; Piaskowska et al., 2021; Sanasi et al., 2023; Lange et al., 2023; Hanifzadeh et al., 2024; Zaiko, 2017).

While the literature, e.g. Klotins et al. (2019), addresses growth, other research discussing this phenomenon (e.g. Unterkalmsteiner et al., 2016; Klotins et al., 2019; Wang et al., 2016) does not use the term ‘scaling’, but instead ‘growth’, particularly in the context of software startups’ growth stage. Zaiko (2017) uses scaling and growth synonymously.

Scaling in business context is a time-limited phase of rapid growth in startups lifecycle, a process of business’ internal transformation (Bohan et al., 2024). Coviello et al. (2024) define ‘scalability’, ‘scaling’, and ‘scale-up’ through a synthesis rooted in organizational scaling. According to Coviello et al. (2024), scaling is an organizational process where management transforms the company using digital resources in a way that outputs expand rapidly without requiring a proportional increase in inputs. This means that scaling increases returns to scale, and not just linearly.

Additionally, Coviello et al. (2024) propose a formula for quantifying scaling:

If $[\text{performance}(t1) / \text{performance}(t0)] / [\text{size}(t1) / \text{size}(t0)]$ exceeds 1, scaling is achieved.

With this formula, scaling can be quantified and measured accurately, and it can be differentiated from generic growth. Scaling is typically associated with rapid expansion, but it can also occur if a company’s size shrinks more slowly than its performance.

Scale-up: Coviello et al. (2024) define a scale-up as an organizational development phase in which the company is undertaking active steps to scale. Companies can move in and out of this phase.

Scalability: Coviello et al. (2024) define scalability as a general organizational capability that is built by aligning and harmonising the company's technology infrastructure, organizational architecture, and business model. Scalability is the foundation for scaling and determines how much a company can scale. Fully digital companies, e.g. Supercell, benefit significantly.

The primary goal of a software startup is to search for a sustainable, scalable, and profitable business model and to grow (Unterkalmsteiner et al., 2016). When successful, these high-growth startups can have a tremendous impact on job markets due to their job creation power (Sanasi et al., 2023).

For the company itself, success in scaling means generating 'returns to scale' by transforming the Organization using digital technologies to expand rapidly while maintaining variable costs and inputs (Coviello et al., 2024). Rayport et al. (2023) define this as achieving 'profit-market fit', where the venture can rapidly increase revenue, gaining additional income per new customer while incurring only marginal costs. Furthermore, findings from a large survey indicate that high growth is a reliable predictor of a company's long-term success (Sanasi et al., 2023). Research by Somaya et al. (2024) shows that high scalability has a positive impact on IPO valuation and access to private VC funding. This suggests that highly scalable companies can delay their IPO until capital needs or valuation reach a higher level.

Being successful in scaling rapidly also increases investor attractiveness, supports further revenue growth, enables the company to capture a larger market share, gain first-mover advantage, establish brand recognition, set industry standards, and build entry barriers for future competitors (Lange et al., 2023). Ultimately, startups should prioritise scaling, as the inability to do so is often linked to their failure to remain viable. (Sanasi et al., 2023).

2.2.2 Timing & Readiness

Choosing the right moment to scale is a challenge for startup companies, and it is in their best interest to succeed in doing so. ‘Speed-first’ scaling strategies, e.g. ‘blitzscaling’, promoted by Reid Hoffman, rely on an aggressive approach where the goal is to achieve dominant market presence through rapid market entry at all costs (Hoffman and Yeh, 2018).

It can be argued that early scaling may help startups prevent their core ideas from being copied, as it allows them to enter the market earlier and potentially establish a strong presence before competitors can respond, and thus, reducing the startup’s imitation risk. However, empirical evidence from Lee and Kim’s (2024) research suggests that there is no substantial benefit for startups that scale early. Premature scaling does not improve the chances of a successful exit or increase profitability. Instead, it increases the likelihood of failure, as early scalers often lack the time to invest in experimentation, e.g. A/B testing, market validation, and achieving product-market fit. The research implies that any potential benefits from reduced imitation risks are outweighed by significant downsides, and therefore concludes that companies should avoid early scaling (Lee and Kim, 2024).

Extrapolation, i.e. profitable scaling, relies on meeting several conditions: having a large enough customer base with shared needs and willingness to pay, and a product that is standardized yet unique (Rayport et al., 2023). Additionally, one or more of the following should be in place to trigger the extrapolation phase: a scalable business model, proven revenue model, network and density effects that increase value and virality, and sufficient funding (Rayport et al., 2023). For a SaaS company, steady recurring revenue, strong cash flows, and a capable team indicate that the company is primed for scaling (Page, 2024).

Lee and Kim’s (2024) study implies that companies should first verify their readiness to scale by confirming product-market fit through rigorous experimentation using methods such as A/B testing. Achieving and verifying product-market fit as a prerequisite for scaling is also confirmed by studies such as Sanasi et al. (2024) and Rayport et al. (2023). Sanasi et al. (2023) discuss the scaling prerequisite more broadly in terms of acquiring market validation. According to Sanasi et al. (2023), Series B funding can be seen as a proxy for market validation, indicating readiness to scale.

Empirical evidence from Lee and Kim (2024) shows that timing is a crucial element for startups, as scaling too early significantly increases failure rates. Based on their findings, startups begin scaling on average four years after founding. Scaling within the first year resulted in a 20–40% increased likelihood of failure, which correlated with less experimentation conducted prior to scaling. Furthermore, their research showed that a greater ability to scale early does not lead to earlier scaling. On the contrary, founders with prior entrepreneurial experience or larger founding teams tend to delay scaling compared to their less experienced peers.

2.3 Business Aspects

This section reviews funding; sales, marketing, and expansion; and growth road mapping, experimentation, and business model evolution.

2.3.1 Funding Strategies

Achieving massive and rapid business scaling (MRBS) requires large amounts of capital and is seen as critical for supporting early market entry and securing a strong position. MRBS startups maintain constant dialogue with both new and existing investors to ensure their satisfaction. This helps secure future funding rounds and retain current investor support. Additionally, MRBS startups leverage investors' industry expertise, advice, and extensive networks (Lange et al., 2023).

Access to financial resources in technology-based businesses is a well-identified constraint for business development (Harrison et al., 2004). Isenberg and Lawton (2014) argue that the most dangerous phase for a startup is scaling up, as rapid growth creates sudden and substantial cash demands.

Startups have various funding strategies to choose from. A widely recognized method for funding scaling is venture capital (VC) (Gompers and Lerner, 2001; Zider, 1998; Isenberg and Lawton, 2014; Pawelski, 2023; Harrison et al., 2004; Lahm and Little, 2005; Patel et al., 2011). Venture capital refers to professionally managed and independent funds that invest

in private companies with high growth potential using equity or equity-linked instruments. (Gompers and Lerner, 2001). VC specialises in addressing this financing gap for young, innovative, high-risk, and potentially high-reward startups, which often struggle to secure sufficient funding through traditional channels, such as banks and public markets. VCs provide capital in exchange for equity or equity-linked stakes (Gompers and Lerner, 2001).

Over 80% of VC funds are used to grow the business, particularly companies in the ‘middle of the classical S-curve’, often through capability building in marketing and sales (Zider, 1998). Funding is provided in rounds if startups meet their milestones. The availability of VC money is generally linked to broader market conditions, such as public market performance and IPO valuations. In addition to funding, VC firms often take a seat on the company’s board and provide advisory support (Gompers and Lerner, 2001). The VC cycle typically includes raising funds, investing, monitoring and supporting portfolio companies, and exiting through acquisition or IPO. This cycle is then repeated.

However, scaling via VC money comes with potential downsides from the founder’s perspective, such as high dilution, liquidation preferences, and possible goal misalignment (Pawelski, 2023). Most VCs operate on a ten-year investment horizon: they invest in the first five years and exit in the following five, targeting approximately three to four times cash-on-cash returns (20 to 30 percent IRR) (Pawelski, 2023). Startup founders must accept ownership dilution and strict exit timelines through IPO or acquisition (Isenberg and Lawton, 2014; Pawelski, 2023).

Bootstrapping refers to a set of creative strategies to acquire, use, and manage resources without relying on traditional external funding, such as equity (Harrison et al., 2004; Patel et al., 2011). It has been described as ‘the purest form of entrepreneurship’ (Lahm and Little, 2005), involving tactics such as cost minimisation, customer- or supplier-funded development, leveraging free resources (Harrison et al., 2004; Patel et al., 2011), and exploiting personal networks (Lahm and Little, 2005). Bootstrapping can reduce or delay the need for external capital, help address early-stage funding gaps (Lahm and Little, 2005) and preserve full control over decision-making and equity (Harrison et al., 2004; Patel et al., 2011). It also forces a lean and cost-efficient operating model (Harrison et al., 2004). However, bootstrapping may throttle the pace of scaling (Patel et al., 2011).

Pawelski (2023) further argues that VC funding may be the most viable path to scaling if the goal is to build a large-impact company. He also notes that there are options beyond classic VC or pure bootstrapping. Isenberg and Lawton (2014) suggest combining multiple financing sources, such as bank loans, customers, suppliers, and public funding, and tailoring pitches to each funder's priorities to maintain cash flow and scale sustainably.

ChartMogul (2024) Growth Report outlines that VC-funded SaaS companies tend to scale faster than their bootstrapped counterparts, particularly after reaching 500,000 dollars in annual recurring revenue (ARR), with slightly higher average retention. However, they are also more vulnerable to macroeconomic downturns and tightening funding conditions. Bootstrapped companies, by contrast, grow more steadily and adapt more easily to market shocks. Both funding paths can lead to success, and the best bootstrappers are often able to reach VC-style metrics.

2.3.2 Sales, Marketing & Expansion

SaaS companies adopt enterprise sales-led, inbound marketing-focused, or completely self-service models (Luoma et al., 2012). **Digital Inbound Marketing (DIM)** refers to the use of organic, online methods to reach and convert potential customers. It's about using online channels to attract potential customers with timely, valuable, and search-friendly content. It leverages strategies such as content marketing, search engine optimization and brand communication, and social media marketing (Opreana and Vinerean, 2015).

Product-led growth (PLG) is a core go-to-market (GTM) strategy, as outlined by Alaghband et al. (2023) and similarly by Page (2024), in which the product itself is leveraged to acquire, grow, and retain customers. PLG relies on digital inbound marketing and combines it with methods such as free trials and freemium models with automated user onboarding to allow fast time-to-value. Integrated product analytics provides aggregated insights that product teams can use to optimize the experience. Alaghband et al. (2023) emphasize that establishing and succeeding with a PLG model requires sufficient funding and recommend a cross-functional focus involving engineering, design, and marketing.

To access high-touch, high-value enterprise deals, a PLG strategy alone is typically not enough. Companies often transition to a more hybrid model, for which Alaghband et al. (2023) coin the term **product-led sales (PLS)**. PLS leverages product analytics to generate product-qualified leads for sales. In this approach, the customer is influenced both by product-led factors such as free trials and transparent pricing, and by personal interactions with the sales team.

Expansion strategy: As a SaaS startup secures a steady stream of acquisitions and strong retention, expansion strategies become critical. These may include upselling existing customers, pivoting to new segments, and international expansion. Geographic expansion can be a major driver in achieving ambitious growth goals. To be successful, companies should develop an understanding of the local market, such as its size, competitors, local preferences, and regulations. This involves adjusting marketing and product elements (e.g. language), establishing a local presence through lean satellite offices or local support, optimizing search engine visibility and advertising, and transferring key resources to support new locations. This includes establishing offices and hiring local sales teams (Viking Venture, 2023). Performance can be measured with metrics such as Monthly Recurring Revenue (MRR) and Net Revenue Retention (NRR) (Paddle, n.d.).

MRBS startups typically have strong customer focus and a unique selling proposition (USP) with a value proposition that creates lock-in effects and barriers to competition. They integrate customer feedback into development and continuously evolve the product in response to market changes and customer needs. They also optimize for reducing churn and keeping customers engaged in the ecosystem. MRBS startups leverage strategic acquisitions, secure funding, and use data to generate additional revenue streams in pursuit of market leadership (Lange et al., 2023). They aim to achieve returns to scale by building and planning for scalability, developing a scalable business model, integrating unit economics, and building data capabilities to support data-led decision making. The goal is to generate returns to scale while allocating resources between short- and long-term priorities, and balancing agility with performance (Mula et al., 2024).

2.3.3 Growth Road Mapping, Experimentation and Business Model Evolution

During the opportunity identification stage of scaling, companies craft a scaling roadmap consisting of high-level management growth goals, broken down into smaller, actionable milestones. These are divided among teams, translated into measurable objectives (e.g. user growth and traffic), and further split into tasks to be completed iteratively in sprints. During this period, companies aim to identify new growth opportunities by examining their users and trying to find underserved segments (Mula et al., 2024).

According to Lange et al. (2023), scanning the environment and recognising opportunities includes continuously monitoring changing trends in markets and technology. This is achieved through market, competitor, and cross-industry analysis, as well as market potential evaluation by size, segment, and dynamics. It also includes monitoring customer needs (e.g. through customer analysis) and identifying characteristics of the target customer, such as willingness to pay (Lange et al., 2023). Rayport et al. (2023) note that, according to classical business theories, companies are either exploring or exploiting. Extrapolation represents a stage of profitable scaling between these classical stages, where the company continues to validate its product-market fit and pursues profit-market fit for expansion of market opportunities achieving higher margins and larger revenue opportunities.

Companies iteratively adjust their business models with asset-light structures. MRBS startups typically adopt light-asset business models and continuously test and validate their business model hypotheses, for example through pilot projects, to confirm problem-solution fit and validate product-market fit in pursuit of desirability, feasibility, and profitability (Lange et al., 2023). They constantly challenge the status quo, critically reflect on their business model, and keep monitoring the market, experimenting and exploring new opportunities, and evolving their product offerings (Lange et al., 2023). Sanasi et al.'s (2023) findings highlight that successful startups continue experimenting even after initial market validation, throughout their scaling journey. These startups use experimentation to refine their value delivery mechanisms (e.g. to target new customer segments, optimise channels, and improve customer relationships). They also extend lean startup principles into growth hacking, using experimentation to drive rapid user activation, retention, and expansion.

During the experimentation stage, companies run experiments (e.g. targeting user clusters with specific features), analyse the results (e.g. testing solutions to user engagement obstacles), and revisit their initial strategies (e.g. refining and integrating successful features) (Sanasi et al., 2023).

2.4 Organizational and Process Aspects of Scaling

This section reviews building organizational capacity, agility, and talent; operational excellence and process innovation; and growth road mapping, experimentation, and business model evolution.

2.4.1 Building Organizational Capacity, Agility & Talent

Building an effective and entrepreneurial staff requires leadership and vision. Companies need to attract and build their teams with top talent. They should systematically identify and manage bottlenecks in employee and management competencies. It is important to establish a company structure and management system, along with effective methods to communicate the company's vision and strategy to ensure alignment (Lange et al., 2023).

This is similarly acknowledged by Mula et al. (2024). The associated activities include gaining capacity by hiring specialist leaders and reorganising the organizational hierarchy (e.g. in marketing, finance, and sales). Companies need to define their strategy and culture, and develop ways to clearly communicate and align the Organization. At this stage, founders also need to learn to trust their employees, loosen control, and delegate accountability and decision-making to specialist leaders. The goal is to grow the Organization while balancing autonomy, structure, and company culture (Mula et al., 2024). Rayport et al. (2023) expand on this by emphasising modular Organizations with autonomous teams. Small, autonomous teams can stay agile and adaptive without losing efficiency. This allows for swift reallocation of talent to focus on the most important initiatives (Rayport et al., 2023).

Founders need to adopt a growth mindset, learn their new priorities and activities, and understand how to direct resources. Founders should identify their gaps, gain the required

knowledge, and learn how to leverage other experts. The goal is to transition into a CEO role. Founders need to learn how to balance between delegation and staying connected (Mula et al., 2024).

Additionally, this stage requires ambidexterity from the company. Key enablers include cultural management, where a strong culture helps teams stay focused, aligned, and innovative; and inorganic growth through strategic acquisitions to add capabilities, increase reach, and enhance scalability (Rayport et al., 2023).

2.4.2 Operational Excellence & Process Innovation

To achieve operational excellence and be capable of MRBS, companies must adopt lean and efficient processes which are constantly refined with qualitative and quantitative measurement, data-led decision-making (e.g. defining and measuring key performance indicators (KPIs) and big data integration), and additionally, by automation and by building flexible and scalable infrastructure for better efficiency and performance. Finally, companies need to invest in marketing and sales, and in identifying and evaluating the right channels to promote the product (Lange et al., 2023).

Expanded by Mula et al. (2024), the activities associated are integrating new technologies (e.g. analytics and automation) to improve existing and create new processes, allow seamless cross-collaboration and knowledge sharing between teams, and remove constraints preventing companies from achieving their growth goals. The goal for this priority is to grow their customer base (Mula et al., 2024).

The constraints removal process is suggested by Rayport et al. (2023). Successful companies understand and leverage the conditions crucial for success, set clear goals, and follow rigorous processes to identify and remove constraints, all while balancing innovation and exploration with disciplined execution of core operations (Rayport et al., 2023).

This requires a systematic, rigorous approach to removing internal business model constraints for growth, from setting a clear goal (e.g. x-times the current revenue), identifying the conditions required (e.g. x-times more customers with y-times frequent

purchases), identifying the constraints to achieving those goals, and addressing them one by one, either by removing them or creating workarounds, starting from the largest blocker (Rayport et al., 2023).

2.5 Technology Aspects

As a startup begins scaling, increasing usage and complexity require a shift from ad hoc methods to more mature engineering and management approaches. This section reviews software engineering before and during scaling; technical debt; and scalable technologies and infrastructure.

2.5.1 Software Engineering

Software engineering involves using structured and measurable methods to design, build, operate, and maintain software systems, applying engineering principles to the software domain (IEEE Computer Society, 2024). Software development is a software startup's core activity (Unterkalmsteiner et al., 2016).

Before the growth stage, Klotins et al. (2019) suggest that startups need to have established a certain level in their team, requirements engineering, and project management functions. As team process prerequisites for scaling software engineering practices, startups should have addressed technical and domain knowledge gaps, established communication channels, hired a capable CTO, and built accountability and trust within teams.

For requirements engineering prerequisites, startups need to have established feedback loops with customers to identify, validate, and invalidate priority features. They should manage feature creep through requirements analysis and begin documenting their requirements. At this point, startups should also improve formal project management practices, such as more structured planning and scheduling, setting budgets, and introducing performance metrics

and goals. Once project management practices are in place, startups begin focusing on internal metrics (e.g. userbase, revenue, and customer satisfaction) to measure their success.

As companies scale, business-related issues, monetization, and marketing are the priority and begin to affect product development decisions. The product must meet the needs of a growing and more diverse user base, requiring flexibility, scalability, and reliability. The need for specialized skills increases, making recruitment a higher priority. Larger teams face more challenges with technical debt as the number of artefacts, coordination needs, and overall complexity grow. This requires startups to manage their technical debt before it becomes a barrier to scaling.

Wang et al. (2016) found that the product is the largest challenge for startups. Klotins et al. (2018) provided evidence that unmanaged technical debt is the single biggest reason for startup failure. As outlined by Lange et al. (2023), about 75% of startups do not survive beyond five years, with the majority failing during the scaling phase. This highlights the importance of technical debt management (Klotins et al., 2019).

Improved testing and automation: Manual, informal testing may be sufficient early on, but scaling requires test suites to ensure faster and more reliable releases. Automation becomes crucial, especially given the need for fast product development, which often stems from internal pressure to validate features quickly and leanly. (Klotins et al., 2019)

2.5.2 Technical Debt Management

Technical debt describes the added effort incurred from maintaining or removing sub-optimal solutions in a software product. It is an umbrella term that can refer to debt arising from various sources, such as software code, user manuals, processes, infrastructure, and knowledge sharing. Debt arises due to insufficient skill, ignorance of best practices, oversight, speed, or pragmatism (Klotins et al., 2018).

Technical debt hinders development and reduces product quality, and therefore will eventually need to be dealt with. Klotins et al.'s (2018) empirical evidence shows a strong link between technical debt and startup failure. Their research indicates that “code smells”

correlate most with declining productivity and product quality. Technical debt peaks and has the most impact during scaling, as product usage increases, team size expands significantly, and internal restructuring occurs (Klotins et al., 2018).

Larger teams face more challenges with technical debt as the number of artefacts, coordination needs, and overall complexity grow. Klotins et al. (2018) suggest that technical debt should be addressed before startups begin scaling. Unterkalmsteiner et al. (2016) agree that uncontrolled technical debt can cause severe challenges. Unterkalmsteiner et al. (2016) propose a systematic, ROI-aware approach for technical debt management, in which all debt should be identified and its impact evaluated. The debt should be addressed according to the highest return on investment. This smart resource allocation approach acknowledges that a startup should not abandon its agility and speed in pursuit of product quality.

2.5.3 Scalable Technologies & Infrastructure

Scalable, robust, and adaptable technology infrastructure plays a key role in a startup's ability to drive and support growth and adapt to evolving needs (e.g. changing requirements and a growing user base), as Emphasized by Lange et al. (2023), Paul et al. (2024), and Ajiga et al. (2024). It is a core requirement for scalability (Klotins et al., 2019). To achieve this, companies need to develop scalable software frameworks, which are a combination of architectural principles, best practices, and modern technologies. The methodologies include adopting microservices architectures, cloud computing, DevOps, robust data management, and monitoring, all of which are critical to support scalability (Ajiga et al., 2024). Integrating technology and big data, typically leveraging state-of-the-art cloud solutions, is crucial for companies to automate and standardise their processes. This is central to operational excellence and enables effective scaling (Lange et al., 2023).

The development of scalable software frameworks is built on three key principles. First, modularity: systems designed as interchangeable components are easier to manage and update. Second, flexibility: frameworks should adapt with minimal modifications to newly introduced technologies. Third, performance optimization: improving systems to handle increased load and complexity effectively (Ajiga et al., 2024).

Microservices architectures involve breaking down applications into small, loosely coupled, independent services that interact through API calls. Each microservice is responsible for a specific function and connected to its own datastore. Microservices increase scalability, flexibility, and resilience due to their independence, service-level scaling, and independent development and deployment. Nayim et al. (n.d.) highlight these benefits in their evaluation of microservices versus monoliths but also point out higher costs compared to the monolithic approach. To meet massive scaling requirements, Netflix adopted microservices by breaking their applications into hundreds of services (Ajiga et al., 2024).

Microservices introduce complexity, including data synchronisation and security challenges, which must be accounted for in system design. This includes leveraging service discovery mechanisms (e.g. Consul and Eureka), monitoring tools (e.g. Prometheus and Grafana), and authentication measures (e.g. OAuth and JWT) (Ajiga et al., 2024).

With cloud computing and on-demand infrastructure startups can develop scalable applications cost-effectively. Cloud computing provides flexibility, elasticity, scalability, and cost optimization. With services such as auto-scaling, compute resources can dynamically scale based on demand. Cloud-native technologies such as Docker and Kubernetes simplify the deployment and management of environments. Docker encapsulates applications and their dependencies, simplifying deployment. Kubernetes orchestrates container deployment and provides features such as load balancing, self-healing, and autoscaling (Ajiga et al., 2024).

DevOps practices are key to developing scalable software frameworks. By promoting collaboration, automating workflows, optimising processes, and fostering a culture of continuous feedback, improvement, and shared responsibility, DevOps allows businesses to adapt to evolving market demands in a competitive landscape while maintaining high performance. CI/CD pipelines are a core element of DevOps. By using tools such as Jenkins, GitLab, and Terraform, companies can automate tasks such as code merging, testing, deployment, and cloud infrastructure provisioning, reducing manual effort and minimising errors. Investment in the right tools and technologies, as well as building a culture of communication and ensuring management support, is essential (Ajiga et al., 2024).

Scalable systems are required to manage data efficiently in the face of growing user demands and complex environments. Data must remain accessible, secure, and efficiently processed.

This requires strategies for scalability, availability, and performance. Distributed databases (e.g. MongoDB), horizontal scaling, and sharding help ensure systems can handle massive volumes and diverse types of data. Data availability, ensuring systems remain operational even in failure, can be supported through replication, automated failover, and fault tolerance. Performance can be Optimized using caching (e.g. in-memory caching such as Redis), distributed caching, and reverse proxy caching. Storing frequently accessed data in RAM or at the network edge shortens retrieval time and reduces database load. Load balancing helps distribute traffic evenly across servers to prevent overload. High availability and fault tolerance can also be supported by automated failover, redundant systems, and replication strategies.

Monitoring tools can be used to analyse and optimise database performance, query execution, and system operation. Real-time monitoring with tools such as Prometheus and Grafana provides insight into the current behavior and performance of systems. Tracking metrics (e.g. system load, response time, and resource utilization) using dashboards enables timely adjustments. Historical data analysis is important for predicting future behavior and understanding long-term performance trends. Proactive management and optimization are crucial to identify and mitigate potential bottlenecks (e.g. queries, latency, processing limits). Data-driven, continuous improvement ensures systems remain scalable as demand evolves (Ajiga et al., 2024).

2.6 Synthesis, Research Gaps & Research Questions

This section synthesizes key insights from the literature review to establish a holistic understanding of software startup scaling, focusing specifically on the context of SaaS startups. This section is organised into three subsections: Section 2.6.1 provides a synthesis of the literature across business, organizational, and technology dimensions. Section 2.6.2 summarizes the key insights. Section 2.6.3 addresses the identified research gaps that guide the development of this study's research questions.

2.6.1 Synthesis

Software Startups: Finding a scalable, profitable business model and achieving high growth is the main goal for startups (Ries, 2007; Blank, 2010). Software startups are characterized as new, high-growth, high-failure-rate companies, working without prior experience and limited resources (e.g. people, skills, and funding), with a focus on building software-intensive products (Klotins et al., 2018; Unterkalmsteiner et al., 2016; Wang et al., 2016; Nguyen-Duc et al., 2021; Melegati et al., 2024). Building the product is both the key priority and challenge (Wang et al., 2016). Software startups develop these products with ad hoc practices in short iterations, experimenting and pivoting (Wang et al., 2016; Klotins et al., 2019; Unterkalmsteiner et al., 2016; Melegati et al., 2024; Bosch et al., 2013; Nguyen-Duc et al., 2021), leading to unique challenges such as prioritizing product development and technical debt management (Unterkalmsteiner et al., 2016; Klotins et al., 2019; Wang et al., 2016).

Concept and timing of scaling: ‘Scaling’ as a term carries some ambiguity among scholars, sometimes being used synonymously or interchangeably with ‘growth’. The majority describe scaling as efficient, rapid growth of a customer base and/or returns to scale, a stage of high or “special” growth (Coviello et al., 2024; Mula et al., 2024; Piaskowska et al., 2021; Sanasi et al., 2023; Lange et al., 2023; Hanifzadeh et al., 2024; Zaiko, 2017). Scaling has been defined in recent literature as an organizational process of internal transformation and leveraging digital resources to achieve returns to scale, a time limited phase of a startup’s lifecycle (Bohan et al., 2024; Coviello et al., 2024). Studies such as Unterkalmsteiner et al. (2016), Klotins et al. (2019), and Wang et al. (2016) use the term ‘growth’ rather than ‘scaling’, usually in reference to the software startup growth stage. Zaiko (2017) uses ‘scaling’ and ‘growth’ as synonyms.

Scaling requires careful timing. Scaling too early or failing to scale can lead to a startup’s demise, outweighing any potential benefits. This underlines the importance of validated product-market fit and rigorous experimentation as a prerequisite (Lee and Kim, 2024; Sanasi et al., 2023; Rayport et al., 2023). This contradicts the view of Hoffman and Yeh (2018), who in Blitzscaling argue that speed should be prioritized above all. For a SaaS company, steady recurring revenue, strong cash flows, and a capable team indicate that the company is primed for scaling (Page, 2024).

Business Dimension of Scaling: Software-as-a-Service (SaaS) is a software delivery model in which the provider hosts, maintains, and keeps the software updated and subscribed users access it via internet (Mell and Grance, 2011; Saltan and Smolander, 2021). SaaS operates using a multi-tenant architecture (Bezemer and Zaidman, 2010) and generates recurring revenue through small monthly or annual subscription fees (Luoma et al., 2012). SaaS leverages high automation and inbound and/or product-led growth, often using a no-touch approach and integrating enterprise sales in later growth stages (Luoma et al., 2012; Alaghband et al., 2023; MADX, n.d.). Key metrics include LTV, churn, retention (Skok, n.d.), the CAC/LTV ratio (target 3:1), and net revenue retention (NRR) (Skok, n.d.). Customer success practices are critical (Cespedes and van der Kooij, 2023; Floerecke, 2018).

Scaling requires substantial funding (Harrison et al., 2004; Lange et al., 2023; Rayport et al., 2023). The most recognized strategy for startup funding is venture capital (VC) (Gompers and Lerner, 2001; Zider, 1998; Isenberg and Lawton, 2014; Pawelski, 2023; Harrison et al., 2004; Lahm and Little, 2005; Patel et al., 2011). However, VC is not the only option. ChartMogul (2024) suggests that bootstrapping, while slower and less risky, can be just as effective as a method for financing growth. Isenberg and Lawton (2014) suggest combining multiple financing sources, such as bank loans, customers, suppliers, and public funding, and tailoring pitches to each funder's priorities to maintain cash flow and scale sustainably.

Organizational & Process Dimension of Scaling: From an organizational perspective, scaling requires reorganizing talent and structure, such as identifying skill gaps, establishing formal structures and communication channels, and clarifying the company vision (Mula et al., 2024; Lange et al., 2023; Rayport et al., 2023). During scaling, founders need to adopt a growth mindset, learn their new priorities and activities, and leverage help as they transition to CEOs (Mula et al., 2024). The aim for organizational structure should be leveraging modular autonomous small teams so the company can stay agile, adaptive, and not lose efficiency; train their employees and make sure their skill bottlenecks are addressed (Rayport et al., 2023).

Modular Organizations with Autonomous Teams Rayport et al. (2023) expand on this by emphasizing modular organizations with autonomous teams. Small, autonomous teams can stay agile and adaptive without losing efficiency. This allows for swift reallocation of talent to focus on the most important initiatives (Rayport et al., 2023).

Systematic Talent Management, Companies need to attract and build their teams with top talent. Continuous Post-validation Experimentation Sanasi et al.'s (2023) findings highlight that successful startups continue experimenting even after initial market validation, throughout their scaling journey. These startups use experimentation to refine their value delivery mechanisms (e.g. to target new customer segments, optimize channels, and improve customer relationships). They also extend lean startup principles into growth hacking, using experimentation to drive rapid user activation, retention, and expansion. This also includes cultivating a growth-centric culture with continuous business model experimentation, strategic acquisitions, and market monitoring to evolve offerings (Mula et al., 2024; Lange et al., 2023; Rayport et al., 2023). Operational excellence and process innovation are achieved by embedding data-led, lean processes; defining clear KPIs; and rigorously removing operational constraints (Mula et al., 2024; Lange et al., 2023; Rayport et al., 2023).

Technology Dimension of Scaling: From a software engineering standpoint, the following prerequisites should be in place before scaling: an established feedback loop with customers to identify, validate, and invalidate priority features; formal project management (e.g. methods, tools, metrics); engineering-business alignment; technical debt management; testing; and automation (Klotins et al., 2019). Klotins et al. (2019) suggest that tech debt if not carefully managed before scaling will slow scaling and can eventually lead to startup failure. Unterkalmsteiner et al. (2016) propose instead a systematic, ROI-aware approach for technical debt management, in which all debt should be identified and its impact evaluated. The debt should be addressed according to the highest return on investment.

From a technological standpoint, scalable, robust, and adaptable technology infrastructure is a core requirement for scalability. This includes cloud computing, microservices, DevOps, robust data management, and monitoring. Microservices increase scalability, flexibility, and resilience by breaking down applications into small, loosely coupled, independent services that interact through API calls. DevOps CI/CD pipelines allow businesses to adapt to evolving market demands by fostering a collaboration culture of continuous feedback, improvement, and shared responsibility and are key in workflow automation, process optimization. Monitoring tools can be used to analyze and optimize database performance, query execution, and system operation. Real-time monitoring provides insight into the current behavior and performance of systems. Tracking metrics (e.g. system load, response

time, and resource utilization) using dashboards enables timely adjustments (Paul et al., 2024; Ajiga et al., 2024; Lange et al., 2023).

2.6.2 Summary

The aim of this literature review was to gain a holistic understanding of software startup scaling. First, to gain an understanding of what is known about software startups, their characteristics, lifecycle challenges, and scaling. Second, to expand the research on scaling from the context of software startups to technology and digital startups due to limited coverage, and finally to delve into the more detailed SaaS business model, e.g. overview, marketing and sales strategies literature and technical scalability.

Software startups build and market innovative software products (Klotins et al., 2019), often with a managed, over-the-internet, subscription-based SaaS model (Saltan and Smolander, 2021), in dynamic environments with limited resources. The startup journey has distinct stages (Klotins et al., 2019) with distinct challenges, and building the product is a core challenge (Wang et al., 2016). The requirement to balance resource constraints and speed creates technical debt. During scaling, software startups transform their business to meet their growth rate and market targets (Klotins et al., 2019). Technical debt peaks during scaling as complexity, usage, and time scales grow. Evidence demonstrates a strong link between technical debt and startup failure (Klotins et al., 2018), requiring management (Unterkalmsteiner et al., 2016).

Current startup scaling research has studied the right timing, identified prerequisites, key practices and strategies, and modeled processes during scaling, mostly in digital startup scaling and from established hubs. Companies should avoid premature scaling (Lee and Kim, 2024) and, as prerequisites, require substantial funding and a validated business model (Lange et al., 2023; Sanasi et al., 2023; Rayport et al., 2023). Scaling from the context of organizational studies is defined as an organizational transformation process aimed at returns to scale with digital technologies (Coviello et al., 2024). This includes evolving the founder role to CEO, establishing hierarchy and distributed decision power, specialization, and autonomy (Lange et al., 2023; Mula et al., 2024; Rayport et al., 2023). Alongside

organizational transformation, companies are required to experiment rigorously, maintain business model innovation (e.g. identify new opportunities and refine current pricing models), and at the same time keep validating their product-market fit (Lange et al., 2023; Sanasi et al., 2023; Rayport et al., 2023). They should also develop scalable technology infrastructure (Ajiga et al., 2024; Lange et al., 2023) with automation and analytics integration.

2.6.3 Research Gaps

Although interest in startup scaling has grown in research, two notable limitations remain in the current literature:

1. Generalized focus on tech and digital startups: Much of the existing literature on startup scaling treats digital startups as a homogeneous group, lacking specificity in understanding the distinct scaling processes and challenges faced by Software and SaaS companies.
2. Limited representation of emerging startup ecosystems: Most scaling research is based on companies operating in mature startup ecosystems such as Silicon Valley, with limited representation of emerging and rapidly growing startup ecosystems like Nordics e.g. Finland. Thus, the applicability is not always clear.

This study addresses these gaps by exploring how Finnish SaaS startups approach scaling through qualitative interviews. Using an empirical, inductive approach, it identifies patterns in how these companies navigate the scaling process, highlighting both shared and context-specific practices.

3 Methodology

This chapter presents the research methodology used in this study. To structure the chapter, seven sections are presented. Section 3.1 outlines the research objective and questions. Section 3.2 explains the research design and introduces the Gioia methodology. Section 3.3 describes the data collection process, including sampling, participants, and interview procedures. Section 3.4 details the steps of data analysis. Section 3.5 addresses research quality and trustworthiness. Sections 3.6 and 3.7 cover ethical considerations and methodological limitations.

Use of AI: During this research, AI was used in data processing steps: transcription, translation, and excerpt refining, as well as later in grammatical error and spelling checks of this study. During the use of AI, the data was always anonymous, and the session was private and temporary; nothing was saved or shared with third parties, and each output was carefully moderated by the author. The author retains full responsibility for all AI-generated outputs and their integration into this study. Detailed descriptions of each AI application appear in the Methodology chapter.

3.1 Research Objective

While interest in startup scaling has grown, two key gaps remain in the literature. First, studies often generalize across digital startups, thus overlooking the specific scaling challenges of Software Startups, and SaaS model. Second, most research focuses on mature ecosystems, with limited attention to emerging regions such as Finland and the Nordic context. SaaS startups face distinct sociotechnical challenges, as the complexity, userbase, team sizes grow (Klotins et al., 2019). During Scaling the tech debt peaks having link to startup failure. (Klotins et al., 2018) Similarly, the Finnish startup ecosystem presents unique cultural and market characteristics e.g. small, digital savvy market that shape how scaling is approached (McKinsey, 2024). This study examines scaling the multifaceted business organization transformation process to achieve returns to scale and including business model and organizational and technical architecture aspects e.g. identified by Lange et al, (2023;

Mula et al., (2024; Rayport et al., (2023; Sanasi et al., (2023) examining this subject specifically from the Software Startup, SaaS and Finnish market context.

Objective of this study is to build empirical understanding how scaling is approached in this context of Finnish SaaS Startups. The study is guided by the following main research question:

How do Finnish SaaS startups approach scaling?

To deepen the inquiry, two supporting questions examine internal and external influences:

1. How do Finnish SaaS startups develop and apply internal strategies during scaling?
2. How do external conditions shape the scaling approaches of Finnish SaaS startups?

These questions are examined through qualitative, inductive research using semi-structured interviews with founders and senior leaders, enabling the identification of both shared patterns and context-specific practices.

3.2 Research Design

This section explains the research design and introduces the Gioia methodology.

3.2.1 Overall Approach

This study was conducted with a qualitative, inductive research design to explore how Finnish startups scale their companies. A qualitative method was chosen for its ability to provide a rich, in-depth understanding of complex social phenomena (Miles and Huberman, 1994), allowing for a nuanced exploration of participants' experiences and perspectives of their scaling journey. The inductive nature of the study (Glaser and Strauss, 1967) facilitated the discovery of new concepts and themes emerging directly from those experiences, allowing the 'how' to emerge from the data itself. The study followed the Gioia methodology.

3.2.2 The Gioia Methodology Framework

This research used the Gioia methodology (Gioia et al., 2013), a systematic approach for conducting inductive qualitative research. This methodology was selected for its structured, two-stage coding process, which facilitates the development of grounded theory while ensuring ‘qualitative rigour’ (Gioia et al., 2013, p. 15). The Gioia method’s emphasis on 1st-order coding, using informant-centric terms, and 2nd-order coding, developing researcher-centric concepts, provides a transparent and auditable process for data analysis.

3.3 Data Collection and Participants

For this study, six Finnish SaaS founders and leaders, from active Finnish SaaS Startups with various employee ranges, and established between 2012-2016, shared experiences from their scaling journey via semi-structured interviews. This section presents the sampling strategy, interview participants, and interview procedures.

Participants, Finnish SaaS startup founders, were selected through purposive sampling (Patton, 2015) with the following criteria: being founding members and central to the scaling journey of currently active Finnish SaaS startups with various employee ranges. All selected companies were established between 2012-2016. The candidates were approached via LinkedIn, phone, and/or with the help of networks (e.g. employees of the candidate companies). After the initial interviews, a mix of purposive sampling and snowball method (Patton, 2015) was applied, as the participants helped by suggesting other founders from their networks. The goal was to reach data saturation. The interviews were scheduled iteratively, and the goal was considered achieved when participants began sharing similar insights.

Total of seven highly knowledgeable informants within the Finnish SaaS ecosystem shared their scaling experience for this study. However, one was excluded because as their business model leaned more towards consulting than SaaS, leaving six final participants. The companies were founded between 2012-2016 with varying headcount ranges. The interviewees of these companies consisted of SaaS founders and early hire Senior Executive.

Table 1 presents an overview of the interview participants, including their roles, founding periods, and company size.

Table 1. Interviewee metadata.

Pseudonym	Founding Period	Headcount Range	Interviewee Role
Int 1	2012 – 2016	501 – 1000	Founder
Int 2	2012 – 2016	101 – 250	Founder
Int 3	2012 – 2016	11 – 50	Founder
Int 4	2012 – 2016	251 – 500	Founder
Int 5	2012 – 2016	11 – 50	Senior Exec.
Int 6	2012 – 2016	501 – 1000	Founder

Founding years and headcount data retrieved from Crunchbase (2025); founding years are aggregated for confidentiality.

The interviews were carried out by Semi-structured interviews (Lim, 2024), conducted via Microsoft Teams. Duration: 45–60 minutes each. Iterative Adaptation: The interview guide was updated as new themes emerged, reflecting the inductive nature of the study (Gioia et al., 2013). Recording and Transcription: With participant consent, all interviews were recorded and transcribed automatically, then verified manually.

3.4 Data Preparation and Data Analysis

The data analysis was conducted as an iterative process, consisting of data preparation, coding, comparison, grouping, and theming.

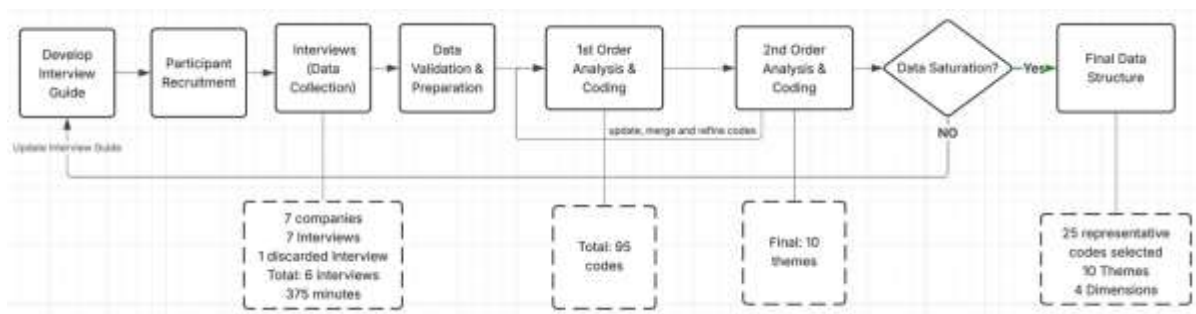


Figure 1. Data Analysis Process

Figure 1. displays the process and stages of this research methodology: Interview guide development; participant recruitment; data collection; data preparation; and systematic and iterative 2-stage analysis and coding. The process was iterated until data saturation achieved.

The auto-generated text-transcripts were carefully corrected by comparing them against the interview video-recordings, one phrase at a time. Simultaneously, all identifying details were removed, and participants were assigned pseudonyms (e.g. Int 3). After the careful data-validation, correction and anonymisation process, the interviews were translated with large-language-model from Finnish to English. The translation was again carefully compared against original source, and manually verified and refined for best possible accuracy. The verified translations of the anonymized transcripts served as the primary data. Data was stored in a secure, private cloud folder and uploaded to the data analysis tool ATLAS.ti.

The data analysis followed Gioia et al. (2013) with systematic, coding of interview insights into concepts, themes and dimensions. It was iterative process with the following main stages:

1. Manual coding of the data

During this stage the data was carefully analyzed and immersed line by line with Atlas.ti data-analysis tool, and manually coded. During this phase the most obvious themes started to emerge. Initially 95 codes was identified.

2. Excerpt refining with large language model

During this stage, a large language model was used to extract and parse representative quotes from the anonymised interviews based on the identified codes to be later displayed in findings.

3. Code and theme merging and Iteration

During this stage, the identified codes and themes were merged and refined iteratively. Some themes were discarded not being strong enough. This left 10 themes.

4. Forming the data structure

Visual data model, helped throughout the analysis process to further re-grouping and re-arranging and merging codes and themes. Final Data structure consisted of 10 themes and of 4 dimensions. 25 representative codes were selected for data structure.

3.5 Research Quality and Trustworthiness

The trustworthiness and quality of this study was ensured by closely following the Gioia methodology (Gioia et al., 2013). During every step of the data analysis, the concepts were reviewed alongside the excerpts they represented to ensure correct grounding in the data. These excerpts are presented in the findings. Additionally, the data structure provides an audit trail showing how insights were formed.

3.6 Ethical considerations

All participants were informed about the study's aims and gave consent to record interviews. Identities were kept anonymous using pseudonyms (e.g. Int 1). Anonymised transcripts remain stored on secure, university-provided cloud storage. The interview recordings have been deleted. Participants were given the option to request a review, edit, or delete their responses from the transcripts. No personal data beyond professional roles was collected, ensuring compliance with general data protection.

Ethical use of AI: All times when AI was used during this research, interview-translation; code and excerpt refinement; and with final spelling-error and grammatic check, The data was always anonymous, and the session was private and temporary, nothing was saved or shared with the third-party.

3.7 Limitations

Despite the rich, in-depth insights this study was able to gain, several limitations need to be recognized. First, reliance on a small sample ($n = 6$), which, even though data saturation was reached, may constrain the generalizability of these findings beyond the examined scope of active Finnish SaaS startups demonstrating successful scaling. As all companies had been established between 2012 and 2016, companies established earlier or later, or those emerging from more heterogeneous backgrounds, could exhibit different scaling patterns. Second, purposive sampling allowed the inclusion of interviewees with data-rich scaling stories but might itself introduce selection bias, prioritizing “success stories” over a more representative cross-section. Third, although structured coding of the Gioia method was applied, qualitative analysis carries the researcher’s perspective; the researcher’s interpretations and the themes emphasized can reflect that subjectivity. Finally, while the use of automated tools to transcribe and later translate several interviews was systematic and correctness was carefully evaluated, minor subtle shifts in nuance may have slipped through.

4 Findings

This chapter presents the key findings of the study from interviews with six Finnish SaaS startups. The findings are organised into scaling prerequisites and 3 core dimensions: business, technology, and organizational following the structure of the literature review. Section 4.1 outlines key prerequisites for scaling, including global ambition, product-market fit, and funding approaches. Section 4.2 presents business strategies related to internationalisation and go-to-market methods. Section 4.3 focuses on organisational and process-related practices. Section 4.4 covers technology-related strategies, including product architecture and technical debt management.

4.1 Scaling Prerequisites

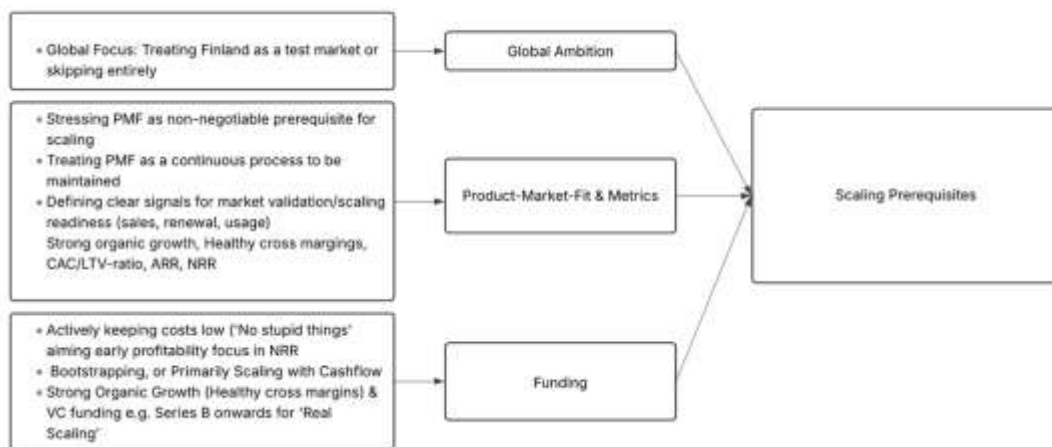


Figure 2. Scaling Prerequisites

Figure 2 displays themes Global Ambition, Product-Market-Fit & Metrics and Funding from dimension Scaling Prerequisites.

4.1.1 Global Ambition

Finnish SaaS companies plan for global markets very early. They tend to treat their domestic market more like a test market to validate their MVP, or may often skip their home market entirely. The small local market is a well-identified constraint.

Some of the Finnish SaaS companies approached global markets by entering them in similarity sequence:

- Int 1: “When we had a budding product-market fit in Finland, we immediately went to Sweden to validate that it wasn't just a local PMF, but to confirm if it was global PMF.”
- Int 2: “First wanted to see if the idea works... if Finland works, try Sweden... then beyond...” Others skip Finland entirely, as in:

Others skipped Finland and targeted global markets early:

- Int 6: “Unlike Nordic firms thinking Sweden/Estonia is international... our first customer was from Mexico... Finland has always been <1% of our sales... the product had to compete globally.”
- Int 3: “Our TAM and customer segment is fully global... The market problem is global... the need is the same in all countries...”
- Int 4: “We actually didn't start in Finland... had a non-compete... which was really good for us... We wanted to be international... everywhere.” These statements show ambition and mindset. This orientation guides their product design approach.

4.1.2 Product-Market Fit & Metrics

Finnish SaaS companies emphasized the importance of having strong product-market fit before scaling, considering strong sales as the main indicator.

- Int 1: "Before even thinking about scaling, you need product-market fit... Many firms miss this point... It's useless to think about scaling before you have PMF."
- Int 3: "Advice: Find the real numbers... dare to face realism through data before concluding readiness to scale..."
- Int 5: "Advice: Validate before scaling... don't scale things that don't work."
- Int 2: "[Readiness indicators:] Can others besides the founders sell it systematically?... Do customers renew?..." These views underscore that it's not enough to find product-market fit once.
- Int 1: "Product-market fit isn't a one-time thing... you have to continuously work to maintain it... PMF runs alongside scaling the whole time." It needs to be continuously re-validated.

Finnish SaaS companies relied heavily on metrics to guide decisions, validate progress, and prioritise work. They accepted that not everything can or should be done at once.

- Int 6: "There are challenges while scaling everywhere [e.g. people, processes, communication] when you grow fast. You can't solve all at once. We focused on the biggest pains."
- Int 3: "Advice: Find the real numbers... dare to face realism through data before concluding readiness... Verify the feeling with math..."
- Int 6: "We logged all feature requests and prioritised them by total revenue potential." Funnel metrics in product-led growth (PLG) and NRR were central to their decision-making.

Finnish SaaS companies followed metrics e.g. ARR, NRR and CAC to LTV ratio built PLG funnels.

- Int 3: "[PLG metrics:] Visitors → Signup% → Activation% → Upgrade% → Retention/NRR → LTV:CAC..."
- Int 5: "Basic KPIs are revenue, ARR..."
- Int 3: "CAC per LTV is the most essential metric... if it meets benchmarks (e.g. 1:5), investors say 'this scales'."
- Int 1: "One of the best proofs of PMF is how you accumulate revenue and at what rate... key figures like retention... Net Revenue Retention."

4.1.3 Funding

All Finnish SaaS companies showcased a somewhat pragmatic approach to funding and spending. Many of them started with very little external capital or were entirely bootstrapped. These companies targeted early profitability to finance their scaling efforts through cash flow.

- Int 1: "We had investors... but needed only €600k to become cash-flow positive... to reach €100M ARR..." Int 5: "Had seed funding behind us... but could say we're quite bootstrapped... funding hasn't been a huge growth driver."
- Int 2: "Company was scaled purely with cash flow... we needed satisfied customers who stay, so we have money to grow operations."

These statements demonstrate a bootstrapped approach or mentality. The companies closely tracked their financial health, targeted revenue and kept their costs down.

- Int 1: "We became profitable relatively quickly... a combination of things: a) we kept costs very low the whole time, we didn't do anything stupid."

Some Finnish SaaS companies raised external funding throughout their scaling journey.

- Int 6: "Our gross margins have always been very healthy... tried to keep over 70%... We grew 3-4x a year purely organically, before raising Series B for intentional scaling." Cost-conscious and efficient use of funds was also evident:

4.2 Business



Figure 3. Business

Figure 3 displays themes Internationalization and Marketing & Sales Strategies from Business dimension.

4.2.1 Internationalization

Finnish SaaS companies establish local hubs or subsidiaries in their key markets. They do this to build local customer-facing teams, tackle time zone differences, build customer trust, and gain access to local market expertise.

- Int 6: "The most important added value was when a customer called, they heard an American accent."

The sequence for setting up these hubs often follows either similarity (Finland → Nordics → English-speaking markets), or is strategically based on data, large market potential, or partnerships.

- Int 1: "We started in Finland, then Sweden, then Germany. Each step ensured it wasn't just a local phenomenon."
- Int 2: "Once we succeeded in Finland, we tried Sweden, then other Nordics, then the Netherlands, US."

- Int 3: "Focus came from organic demand/SEO... Will recruit first salespeople into regions with best self-serve traction... Data largely dictates where we go (US, Central Europe)."
- Int 1: "[Hub identification:] Where were the hubs with our potential customers?... Became clear our ICP was in places like Berlin, London..."

4.2.2 Marketing & Sales Strategies

Finnish SaaS companies leveraged inbound marketing and digital channels (e.g. social media, SEO, and paid ads) to reach potential customers and achieve global reach. Some initially relied solely on product-led, self-service growth, while others supported this with inbound sales teams.

- Int 6: "First five years... pure inbound sales... Customer comes to us having almost decided to buy... our job is to answer questions and confirm it's a damn good decision." Int 5: "Our main channel now is inbound."
- Int 3: "Pure global self-service model. Idea is infinite scale without human effort... pure credit card plans, self-service users."
- Int 4: "It was completely a product-led growth model at the start... Product was built from the start so the customer could self-serve onboard, pay... didn't necessarily need any contact with us."

Some of the Finnish SaaS companies leveraged growth hacking techniques such as targeted competitor social media follower for cost efficient and targeted customer reach:

- Int 6: "[First customers] found via Twitter ads. We dug up followers of all our competitors... gave very targeted marketing."

Others relied on platform partner referrals, creating win-win-win scenarios by solving the problems of their partners' most demanding customers:

- Int 1: "We helped our partner with their most demanding customers... started to get intros... used these as lighthouse cases..."

One company, with a more complex product requiring market-by-market modification, relied on an enterprise sales model from the beginning. Others described adopting the enterprise model later in their scaling journey for high-touch deals.

- Int 6: "Tested outbound during B round... but it wasn't a goldmine then... C and D [rounds] have been strongly about building a larger outbound sales machine... Until then, could grow quite well with pure inbound."

4.3 Organizational & Process

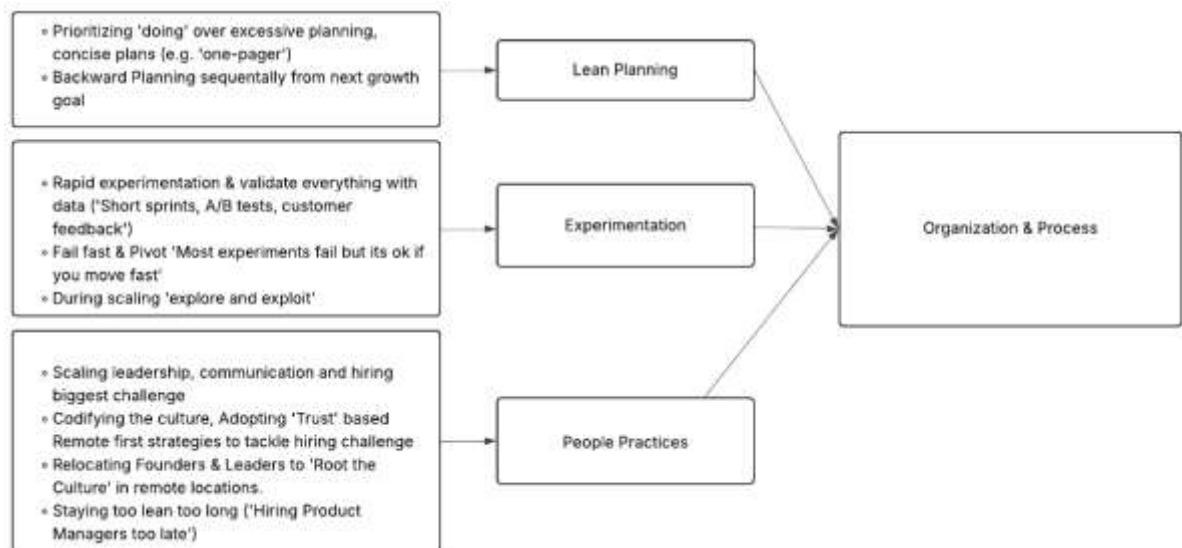


Figure 4. Organization & Process

Figure 4 displays themes Lean Planning, Experimentation and People Practices from Organization & Process dimension.

4.3.1 Lean Planning

Finnish SaaS companies prioritised lightweight planning, execution, learning by doing, and pivoting.

- Int 1: "We had a plan, but it was quite 'light'... maybe one A4 page for entering Germany... focus was 'let's go and get customers... reach this point, then see what's next'."
- Int 4: "Using extremely concise plans (e.g. 'one-pager')."
- Int 2: "Decided and then went for it... requires clear decision everyone commits to, then try full force... easy to plan endlessly."

Best practice planning philosophies including:

- Int 6: "The most sensible way I've ever seen to plan is that you consider where you want to be in a couple of years, and from there you look backward... As a planning method, it's superior compared to the incremental approach."

4.3.2 Experimentation

Finnish SaaS companies demonstrated the use of experimentation, learning, and adapting based on their successes and failures. They conducted experiments and collected data through customer interactions or from the product via A/B testing, usage tracking, and end-user interviews.

- Int 3: "Iteration in ~2-week sprints... looked at numbers... Usually, nothing happened [immediately]... As long as we didn't go backwards, that was most important... Sometimes we did... adding onboarding steps hurt activation, had to quickly remove."
- Int 6: "Tried all kinds of things, most failed. Not the end of the world as long as you can move and implement quickly."
- Int 3: "We only bother the software developer once we are 100% sure that something works... First, validate the idea with Figma, test with customers and stakeholders... And prioritise the features with most revenue potential."

- Int 2: "Testing was mostly: 'Let's meet 100 potential customers and see what happens'... exposing the software to users [via sales] was the best market research."

They also structured and reorganised talent based on skillsets, acknowledging that some people are better at exploring, while others excel at optimising.

- Int 1: "Different people for Explore vs. Exploit: One type validates new things quickly... experiments... builds fast. The other optimises something existing."

4.3.3 People Practices

Some Finnish SaaS companies described establishing early remote and asynchronous development teams and adopting remote-first strategies.

- Int 4: "We were early adopters of remote work... our first developer was from Poland." The local market and developer pool were seen as constraints, which global hiring helped to overcome.
- Int 6: "After COVID, we started hiring remotely in different countries... eventually devs worldwide." Trust was consistently highlighted as a cornerstone, particularly in enabling autonomy and distributed teams.

Remote hiring allowed access to larger talent pool but required mutual trust and adjustments in hiring and culture practices.

- Int 6: "[Future team structure:] Maintenance/DevOps team + 'Attack' team for new features." Adapting to remote-first practices required changes in communication and decision-making strategies.
- Int 4: "Forced ourselves into the remote model... agreed no decisions made that aren't visible to everyone. Slack, daily video calls, everything was documented so remote people wouldn't be left out... The cornerstone is trust "

Some Finnish companies described strategies for “scaling their culture” such as relocating their founders to establish the remote offices and or codifying the culture.

- Int 6: "Major sites... always had someone from the company [early on]... Culture came via osmosis... More problems in places that were purely remote..." Distributed teams required codifying the culture.

4.4 Technology

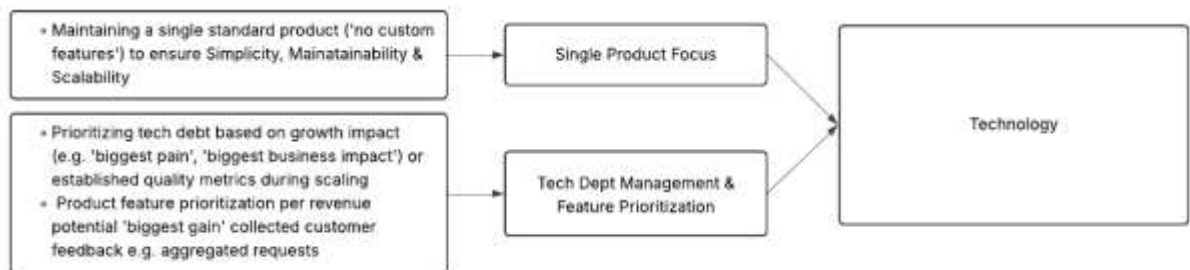


Figure 5. Technology

Figure 5 displays themes Single Product Focus, Tech Dept Management & Feature Prioritization from Technology Dimension.

4.4.1 Single Product Focus

All Finnish SaaS companies Emphasized the importance of a single product to maintain a scalable and maintainable solution that can grow globally. Most began with a single product in English only, supporting their globalisation strategies. The Nordics often acted as an entry market, and more languages were added in later scaling stages.

- Int 4: "We deploy continuously, and everyone's on the same codebase, no custom forks even for big clients. It's simpler that way."
- Int 6: "We've always stuck to a single product strategy. Sometimes a customer might be a pilot user for a new feature, but then we roll it out for everyone."

4.4.2 Technical Debt Management Strategies

The Finnish SaaS companies developed technical debt management strategies and pragmatically navigated them during scaling, prioritising the biggest growth blockers and enablers at each stage. Some engineer-led, VC-funded companies built initially more scalable products.

- Int 6: "Unlike sensible people who would build a very minimal MVP, we took an engineer-driven approach and spent about a year coding, but it actually de-risked a lot for us by solving a lot of the future scalability issues."

Others, mainly cash-flow-funded companies prioritised speed and product-market fit, validating their MVPs with real users.

- Int 1: "We incurred technical debt early, optimising for speed to find PMF."
- Int 2: "We stayed lean, focusing on core features done well." This helped reduce the risk of building the wrong product and allowed them to adapt quickly to market demands.

Some companies acknowledged technical debt as a challenge:

- Int 2: "Tech debt slowed us down."

However, all companies had established prioritization methods, focusing on business impact. Prioritization was driven by the biggest obstacles to growth, customer pain points or gains, and quality-related metrics.

- Int 1: "[Prioritization was based on] where are the biggest obstacles to growth right now?"
- Int 6: "[Used metric] 'problems per 1000 nodes,' aiming to decrease it via automation... Good proxy for quality."

5 Discussion

This study of Finnish SaaS startups' scaling identified strategies and practices that align with existing empirical understanding while also highlighting unique, context-specific characteristics.

Unique characteristics include:

- Large funding was not a prerequisite for scaling in most of these Finnish SaaS companies. Instead, it served to support scaling, providing variance from the findings of studies such as Lange et al. (2023) and Rayport et al. (2023).
- All startups planned early for global markets. It was not common to stay as “regional champion” mentioned in the McKinsey report (Bjørndalen et al., 2024). Globalization was approached either immediately or sequentially.
- Technical debt was managed not prior to, but during scaling, based on maximum impact demonstrating a varied approach from that presented by Klotins et al. (2019).

These startups navigate scaling through lean planning, agile experimentation (Sanasi et al. 2023), validating their decisions with data (Lange et al., 2023) while ensuring strong PMF. Their product aims for simplicity, with non-negotiable single product focus.

This chapter discusses how Finnish SaaS startups approach scaling based on the empirical findings. The discussion is structured around the supporting research questions. Section 5.1 focuses on internal strategies developed and applied by Finnish SaaS startups during scaling, while Section 5.2 focuses how external conditions shape those strategies. The discussion connects insights from the empirical data with relevant literature to highlight both alignment and discuss context unique approaches.

5.1 How do Finnish SaaS startups develop and apply internal strategies during scaling?

Finnish SaaS companies keep their plans short and aim for lean practices but validate their decisions carefully with data. They have a strong culture of experimentation, learning both from their mistakes and successes, and focus pragmatically on the main priorities.

Finnish SaaS startups treat product-market fit (PMF) as a non-negotiable requirement and prerequisite before scaling. They aim to validate and continue revalidating it as they scale and expand into new markets. This aligns with research by Rayport et al. (2023) and Sanasi et al. (2023). For validation, Finnish companies rely purely on financial metrics. The most genuine proof is revenue from actual users, which Rayport et al. (2023) also emphasises. This careful, data-led and cost-conscious validation was clearly demonstrated.

Another non-negotiable decision is their single product strategy. Finnish SaaS companies demonstrate simplicity and practicality. If possible, the product is the same for everyone, with one codebase and no custom work or exceptions. This ensures scalability and maintainability.

Finnish SaaS companies focus on a limited set of features at a time, based on the most revenue gained through aggregated feedback or customer testing. This same thinking also applies to how they handle technical debt. Finnish SaaS startups describe a technical debt management strategy that somewhat diverges from Klotins et al. (2019). Instead of aiming to manage debt pre-emptively before scaling, they deal with it pragmatically during scaling, prioritising the biggest blockers at hand. They rely on product quality metrics or prioritise based on the greatest business impact by revenue. Some highlighted that technical debt did slow their scaling, but most described similar strategies.

5.2 How do external conditions shape the scaling approaches of Finnish SaaS startups?

The small Finnish local market is a well-identified constraint among the respondents. Finnish SaaS companies aim for global markets from the beginning. Their approach follows either the Uppsala model by Johanson and Vahlne (2009), advancing in sequence (Finland → Sweden → Nordics → other English-speaking countries → rest of the world), or they often bypass their home market entirely and aim for global markets from the outset, following the Born Global model described by Rummyantseva and Welch (2023).

This sequential approach was more common among Finnish SaaS companies targeting profitability and sustainable scaling, and the sequence was often due to market similarity. In contrast, the initially global approach was dominant among companies that had raised more venture capital.

The recent report from McKinsey (Bjørndalen et al., 2024) did not address this form of sequential, more traditional Uppsala-model-style globalisation. Instead, it categorised Nordic software companies as either global from inception or regional champions, potentially suggesting that this sequential approach to global markets could be a more unique strategy among Finnish SaaS companies.

Finland was treated as a test market by some Finnish SaaS companies. In Finland, the market does not expect local language support, especially for software. This gives Finnish SaaS companies the opportunity to build one version of the product targeting all English-speaking markets, while still advancing market by market and using their domestic market as an MVP testbed.

While research on scaling (e.g. Rayport et al., 2023; Lange et al., 2023) suggests that substantial funding, typically in the form of venture capital, is a critical requirement for scaling, the findings here demonstrate divergence. Half of the startups aimed for early profitability and scaled using their own cash flow, validating their strategies carefully with data. While these startups did raise funding at some point during their scaling journey, it was often to accelerate growth. Some bootstrapped entirely.

Existing research already recognises bootstrapping as a viable path to scale, and the data in this study highlights it as a common funding approach among globally successful Finnish SaaS startups. Although the findings do not directly explain why Finnish companies take this route, it is known that less venture capital is available in the Nordic region. To scale globally without VC backing, the only viable approach may be through sustainable, careful growth. These companies were very cautious, or “frugal”. with their resources, aiming to make smart decisions, such as collecting customer data and experimenting carefully before allocating engineering effort. This may suggest a contextual cultural factor, often associated with Finnish mentality, or simply a common practice shared among successful Finnish companies.

There was a clear divergence in how initial MVPs and early products were built. The Lean Startup method by Ries (2007), popular among Silicon Valley startups, was used by only half of the Finnish SaaS companies. The other half, consisting mainly of engineer-led founder teams, tended to build more robust, over-engineered MVPs, with few or no pivots, intentionally delaying time to market. This “engineering mindset” and the stark divergence could be due to the high proportion of engineers in Finland.

Another dominant practice among Finnish SaaS companies was the use of inbound marketing, either through a self-service, product-led growth model, or supported by inbound sales teams. They employed cost-efficient techniques for early market validation, including digital growth hacking methods. After initial validation, companies established hubs in key markets to build customer trust, address time zone challenges, and gain local market expertise. Later in their scaling journeys, companies adopted hybrid sales strategies, including enterprise sales models for high-touch engagements. Finally, some Finnish companies were early adopters of trust-based, remote-first strategies. These included codified culture, asynchronous communication tools, and routine video meetings. These strategies, built on trust, enabled them to establish effective distributed development teams and access global talent pools, addressing the limitations of the small local job market.

6 Conclusions

This study explored the scaling processes of Finnish SaaS startups through the lived experiences of founders and leaders. It was guided by the main research question: How do Finnish SaaS startups approach scaling? The goal was to build an empirical understanding of scaling approaches in the context of Finnish SaaS startups while identifying both shared strategies and context-specific adaptations.

This study identified and described internal strategies emerged of Finnish SaaS founders' scaling journeys that align with current digital startup studies for major startup hubs, e.g., lean planning, strong experimentation culture, single-product strategy, customer-feedback integration and careful, data-validated decision-making. Additionally, it highlights a technical-debt management approach, suggesting a divergent, potentially context-specific strategy.

It also identified strategies and approaches shaped by external conditions of a small local market: introducing global ambition and early globalization; abundant engineering talent, which Emphasized engineer-founder teams and scaling practices; digital maturity, e.g., leveraging Finland as a potential test market and building single English-language interfaces. Additionally, Finnish SaaS companies leveraged a global talent pool to tackle limitations of the small home market by adopting early, trust-based remote strategies. Further, the study identified a potentially context-related internationalization approach divergent to other similar characteristic Nordic companies to complement the McKinsey report (Bjørndalen et al., 2024). All the companies targeted global markets. No company planned initially Nordic only. For those funding their internalization with operational cashflow the expansion happened market by market in similarity sequence with early profitability. Companies with external funding and PLG strategies targeted immediately global markets.

This research contributes to the literature by offering a contextualised view of scaling in the Finnish SaaS landscape. It identifies which strategies similar to those in earlier research and models of scaling are employed, and at the same time how geographical context potentially shapes those strategies. It also identifies some distinct characteristics of Finnish SaaS scaling, even diverging from the Nordic context.

For practitioners, especially Finland-based current and future founders, the following takeaways are suggested:

- Critical requirement prior scaling: Validate Product Market Fit and Revenue Model
- Bootstrapping is viable approach for global scale with systematic data-led, resource efficient decision making, systematic market by market expansion
- Product-led growth and inbound marketing allow immediate global reach
- Establishing local hubs & teams in key markets tackle trust and time zone issues and gain access to local market insights
- Set incremental growth targets and backtrack for organizational requirements
- Identify, prioritise and systematically remove key constraints blocking business growth

For future research:

1. This study focused on scaling journeys of successful Finnish SaaS startups; to provide a richer understanding of scaling in this context, future research could examine how unsuccessful Finnish SaaS startups approach scaling.
2. This study uncovered both shared practices and divergence in scaling strategies compared to research on more established startup hubs; similarly, it could be valuable to investigate how other emerging startup hubs approach scaling.

Scaling is a critical but challenging stage for software startups, with most failing at this point (Sanasi et al., 2023). This study explored scaling in the Finnish SaaS context, identifying the core strategies these companies use to grow successfully from a small, digitally mature market. These findings contribute to understanding of startup scaling in smaller markets and call for further empirical research for this context unique process.

References

Ajiga, D. (2024). Methodologies for developing scalable software frameworks that support growing business needs. 10.51594/ijmer.v6i8.1413.

Alaghband, M., Panagiotidou, N., Roche, P. and Schneider, J. (2023) From Product-led Growth to Product-led Sales: Beyond the PLG Hype. McKinsey & Company, August. Available at: <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/from-product-led-growth-to-product-led-sales-beyond-the-plg-hype> (Accessed: 1 November 24).

Bezemer, C. and Zaidman, A. (2010) ‘Multi-tenant SaaS applications: maintenance dream or nightmare?’, in Proceedings of the Joint ERCIM Workshop on Software Evolution (EVOL) and International Workshop on Principles of Software Evolution (IWPSE) (IWPSE-EVOL '10). New York: Association for Computing Machinery, pp. 88–92. doi: 10.1145/1862372.1862393.

Bjørndalen, A.J., Dahlström, P., Lundberg, T. & Torres, A. (2024) ‘What’s driving the Nordic countries’ software export surge?’, McKinsey & Company. Available at: <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/whats-driving-the-nordic-countries-software-export-surge> (Accessed: 22 April 2025).

Blank, S. (2010) ‘What’s a startup? First principles’, Steve Blank Blog, 25 January. Available at: <https://steveblank.com/2010/01/25/whats-a-startup-first-principles/> (Accessed: 25 January 2025).

Bohan, S., Tippmann, E., Levie, J., Igoe, J. & Bowers, B. (2024) ‘What is scaling?’, Journal of Business Venturing, 39(1), p.106355. doi:10.1016/j.jbusvent.2023.106355.

Bosch, J. & Olsson, H. & Björk, J. & Ljungblad, J.. (2013). The Early Stage Software Startup Development Model: A Framework for Operationalizing Lean Principles in Software Startups. LESS 2013. LNBIP. 167. 1-15. 10.1007/978-3-642-44930-7_1.

Cespedes, F.V. & van der Kooij, J. (2023) ‘The Rebirth of Software as a Service’, Harvard Business Review, 18 April. Available at: <https://hbr.org/2023/04/the-rebirth-of-software-as-a-service> (Accessed: 20 January 2025).

ChartMogul (2024) SaaS Growth Report: Benchmarking Growth Trends from Bootstrapped and VC-backed SaaS Businesses. Available at: <https://chartmogul.com/reports/saas-growth-vc-bootstrapped/> (Accessed: 25 January 25).

Crowne, M.. (2002). Why software product startups fail and what to do about it. Evolution of software product development in startup companies. Caring for The Ages. 1. 338 - 343 vol.1. 10.1109/IEMC.2002.1038454.

Floerecke, S. (2018). Success Factors of SaaS Providers’ Business Models – An Exploratory Multiple-Case Study. 10.1007/978-3-030-00713-3_15.

Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. *Organizational Research Methods*, 16(1), 15–31. <https://doi.org/10.1177/1094428112452151>

Glaser, B.G. and Strauss, A.L. (1967) The discovery of grounded theory: strategies for qualitative research. Chicago: Aldine.

Guest, G., Bunce, A. & Johnson, L. (2006) How many interviews are enough? An experiment with data saturation and variability. *Field Methods*, 18(1), 1 pp. 59–82. doi: 10.1177/1525822X05279903.

Harrison, Richard & Mason, Colin & Girlingz, Paul. (2004). Financial bootstrapping and venture development the software industry. *Entrepreneurship and Regional Development - ENTREP REG DEV*. 16. 10.1080/0898562042000263276.

Hanifzadeh, F. & Talebi, K. & J. Sadeghi, V. (2024). Scalability of startups: the impact of entrepreneurial teams. *Journal of Global Entrepreneurship Research*. 14. 10.1007/s40497-024-00383-7.

Hoffman, R. and Yeh, C. (2018). *Blitzscaling: The Lightning-Fast Path to Building Massively Valuable Companies*. Currency, New York.

IEEE Computer Society. (2024) Guide to the Software Engineering Body of Knowledge (SWEBOK), Version 4.0. Available at: <https://ieeecs-media.computer.org/media/education/swebok/swebok-v4.pdf>

Isenberg, D. and Lawton, D. (2014) How to Finance the Scale-Up of Your Company. *Harvard Business Review*, 18 August. Available at: <https://hbr.org/2014/08/how-to-finance-the-scale-up-of-your-company> (Accessed: 25 January 2025).

Johanson, J. & Vahlne, J-E. (2009). The Uppsala Internationalization Process Model Revisited: From Liability of Foreignness to Liability of Outsidership. *Journal of International Business Studies*. 40. 1411-1431. 10.1057/jibs.2009.24.

Klotins, E., Gorschek, T., Lenarduzzi, V., and Regnell, B. (2021). A Progression Model of Software Engineering Goals, Challenges, and Practices in Start-Ups. *IEEE Transactions on Software Engineering*, 47(3), pp. 498-521. doi: 10.1109/TSE.2019.2900213.

Klotins, E., Unterkalmsteiner, M., Chatzipetrou, P., Gorschek, T., Prikladnicki, R., Tripathi, N. and Pompermaier, L. (2018). Exploration of technical debt in start-ups. 75-84. 10.1145/3183519.3183539.

Lahm, R. & Little, H. (2005). *BOOTSTRAPPING BUSINESS START-UPS: A REVIEW OF CURRENT BUSINESS PRACTICES*.

Lange, F., Tomini, N., Brinkmann, F., Kanbach, D.K., & Kraus, S. (2023). Demystifying massive and rapid business scaling – An explorative study on driving factors in digital start-ups. *Technological Forecasting and Social Change*, 196, 122841. Elsevier. <https://doi.org/10.1016/j.techfore.2023.122841>.

Lee, S. & Kim, J.D. (2024). When do startups scale? Large-scale evidence from job postings. *Strategic Management Journal*. 45. 10.1002/smj.3596.

Gompers, P. & Lerner, J. (2001). The Venture Capital Revolution. *Journal of Economic Perspectives*. 15. 145-168. 10.1257/jep.15.2.145.

Lim, W.M., 2024. What is qualitative research? An overview and guidelines. *Australasian Marketing Journal*, [online] Available at: <https://doi.org/10.1177/144135822412646198> (Accessed 17 Jan. 2025).

Luoma, E., Rönkkö, M. & Tyrväinen, P. (2012). Current Software-as-a-Service Business Models: Evidence from Finland. 114. 181-194. 10.1007/978-3-642-30746-1_15.

Mell, P. & Grance, T. (2011) The NIST definition of cloud computing. National Institute of Standards and Technology, Special Publication 800-145.
<http://dx.doi.org/10.6028/NIST.SP.800-145>

Nguyen-Duc, A., Kemell, K.K. & Abrahamsson, P. The entrepreneurial logic of startup software development: A study of 40 software startups. *Empir Software Eng* 26, 91 (2021).
<https://doi.org/10.1007/s10664-021-09987-z>

Mula, C., Zybura, N., & Hipp, T. (2024). From digitalized start-up to scale-up: Opening the black box of scaling in digitalized firms towards a scaling process framework. *Technological Forecasting and Social Change*, 202, 123275. Elsevier.
<https://doi.org/10.1016/j.techfore.2024.123275>.

Miles, M.B. and Huberman, A.M. (1994) *Qualitative data analysis: An expanded sourcebook*. 2nd edn. Thousand Oaks, CA: SAGE.

Opreana, A. & Vinerean, S. (2015). A New Development in Online Marketing: Introducing Digital Inbound Marketing. 3. 29-34.

Paddle (n.d.) The Ultimate Guide to SaaS Expansion. Available at: <https://www.paddle.com/resources/saas-expansion-guide> (Accessed: 25 January 2025).

Page, J. (2024) Top Strategies to Effectively Scale Your SaaS Business. SaaS Academy. 1 October. Available at: <https://www.saasacademy.com/blog/strategies-to-scale-your-saas-business> (Accessed: [date]). MADX (n.d.) Growth in SaaS: Key Strategies for 2025. Available at: <https://www.madx.digital/learn/growth-saas> (Accessed: 25 January 25).

Patel, P. C., Fiet, J. O., & Sohl, J. E. (2011). Mitigating the limited scalability of bootstrapping through strategic alliances to enhance new venture growth. *International Small Business Journal*, 29(5), 421-447. <https://doi.org/10.1177/0266242610396622>

Patton, M.Q. (2015) *Qualitative Research & Evaluation Methods*. 4th edn. Thousand Oaks, CA: SAGE.

Paul, A. & Kelvin, L. & Brown, K. (2024). Optimizing IT Growth: Strategies for Building and Scaling Robust Infrastructure Systems. 10.13140/RG.2.2.25508.24965.

Pawelski, J. (2023) Bootstrap vs VC — The Ultimate Guide For Founders. Medium. Available at: <https://medium.com/@pawelski/bootstrap-vs-vc-the-ultimate-guide-for-founders-9d7ed7af8ec0> (Accessed: 24 January 2024).

Piaskowska, D., Tippmann, E. , Monaghan, S. (2021) Scale-up modes: Profiling activity configurations in scaling strategies, *Long Range Planning*, Volume 54, Issue 6. 102101, ISSN 0024-6301, <https://doi.org/10.1016/j.lrp.2021.102101>. (<https://www.sciencedirect.com/science/article/pii/S0024630121000327>)

Ries, E. (2011). *The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to Create Radically Successful Businesses*. Crown Business, New York.

Rumyantseva, M., Welch, C. The born global and international new venture revisited: An alternative explanation for early and rapid internationalization. *J Int Bus Stud* 54, 1193–1221 (2023).

Saltan, A. & Seffah, A. (2018) Engineering and Business Aspects of SaaS Model Adoption: Insights from a Mapping Study. Available at: https://www.researchgate.net/publication/330398139_Engineering_and_Business_Aspects_of_SaaS_Model_Adoption_Insights_from_a_Mapping_Study

Saltan, A. and Smolander, K. (2021) 'Bridging the state-of-the-art and the state-of-the-practice of SaaS pricing: A multivocal literature review', *Information and Software Technology*, 133, p. 106510. doi: 10.1016/j.infsof.2021.106510.

Sanasi, S., Ghezzi, A., Cavallo, A. (2023) What happens after market validation? Experimentation for scaling in technology-based startups, *Technological Forecasting and Social Change*, Volume 196, 2023, 122839, ISSN 0040-1625, <https://doi.org/10.1016/j.techfore.2023.122839>.

Skok, D. (n.d.) SaaS Metrics 2.0 – A Guide to Measuring and Improving What Matters. For Entrepreneurs. Available at: <https://www.forentrepreneurs.com/saas-metrics-2/> (Accessed: 20 January 2025).

Somaya, D. & You, J. (2024). Scalability, venture capital availability, and unicorns: Evidence from the valuation and timing of IPOs. *Journal of Business Venturing*. 39. 106345. 10.1016/j.jbusvent.2023.106345.

Thomas, D. (2006). A General Inductive Approach for Analyzing Qualitative Evaluation Data. *American Journal of Evaluation*. 27. 237-246. 10.1177/1098214005283748.

Viking Venture (2023) How to Succeed with Geographic Expansion as a B2B SaaS Company. 23 November. Available at: <https://vikingventure.com/how-to-succeed-with-geographic-expansion-as-a-b2b-saas-company/> (Accessed: 1 October 2024).

Unterkaustein, M., Abrahamsson, P., Wang, X., Nguyen-Duc, A., Shah, S., Bajwa, S.S., Baltes, G.H., Conboy, K., Cullina, E., Dennehy, D., Edison, H., Fernandez-Sanchez, C., Garbajosa, J., Gorshek, T., Klotins, E., Hokkanen, L., Kon, F., Lunesu, I., Marchesi, M., Morgan, L., Oivo, M., Selig, C., Seppänen, P., Sweetman, R., Tyrväinen, P., Ungerer, C. and Yagüe, A., 2016. *Software Startups – A Research Agenda*. e-Informatica Software

Engineering Journal, 10(1), pp.89–124. [online] Available at: <https://doi.org/10.5277/e-Inf160105>.

Zajko, M., 2017. Challenges of scaling-up process for start-ups. Balkan Region Conference on Engineering and Business Education, 3(1), pp.62–70. Sciendo. DOI: 10.1515/cplbu-2017-0009

Zider, B. (1998) 'How Venture Capital Works', Harvard Business Review, (November–December). Available at: <https://hbr.org/1998/11/how-venture-capital-works> (Accessed: 5 October)



Appendix A. Gioia Data Structure